Unit 13, Lesson H1: Types of Transformations

Benchmark

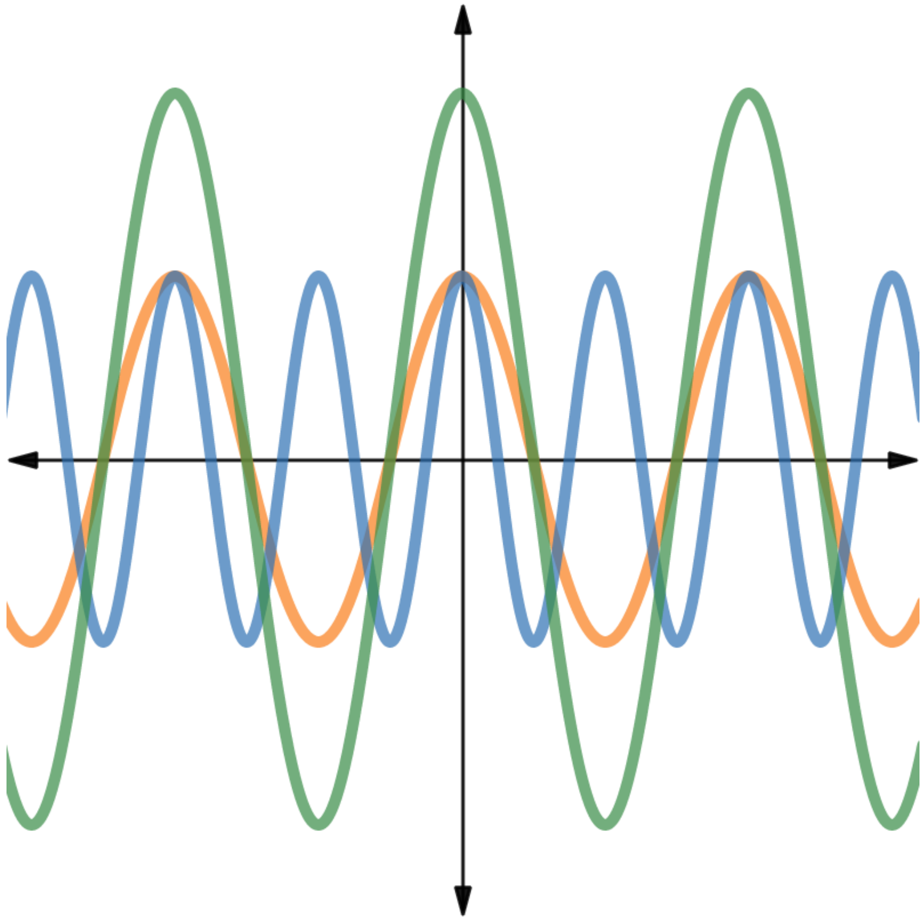
MA.912.F.2.3 Given the graph or table of $f(x)$ and the graph or table of $f(x)+k$, $kf(x)$, $f(kx)$, and $f(x+k)$, state the type of transformation and find the value of the real number k.

Learning Target

☐ I can state the transformation and find the value of the real number k when given the graph or table of $f(x)$ and the graph or table of $f(x)+k$, $kf(x)$, $f(kx)$, and $f(x+k)$.

13.H1.1 Warm-Up: Transformation of a Wave

1. The graph shows three different functions. The **orange** graph represents a parent function. The **blue** and **green** graphs are transformations of the parent function.



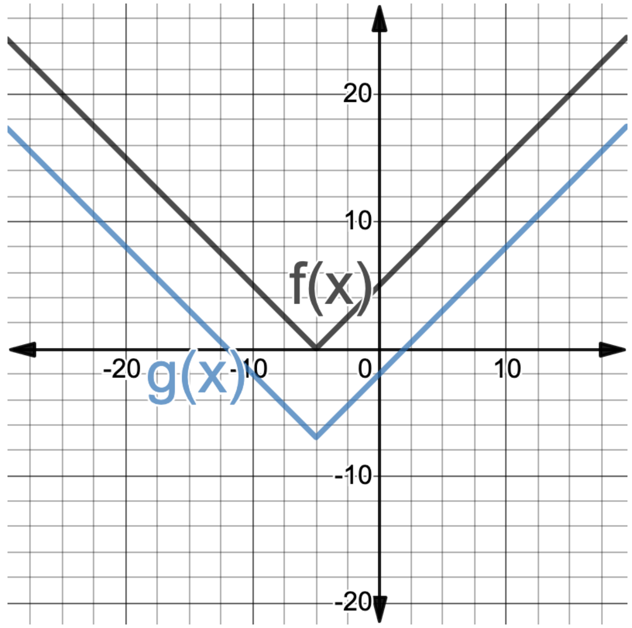
Terrence and Camila described the graphs.

Terrance	Camila
The green graph is taller than the orange graph, so the y-values were multiplied. The blue graph has more waves than the orange graph and there are more x-intercepts, so the x-values were divided.	The green graph is a stretch of the orange graph vertically, so the y-values were multiplied. The blue graph is a compression horizontally of the orange graph, so the x-values were divided.

Whose description is closest to how you would describe the graphs?

13.H1.2 Guided Instruction: Describing Transformations

1. The functions $f(x)$ and $g(x)$ are shown on the graph.



Work with your partner to complete the table to describe $g(x)$.

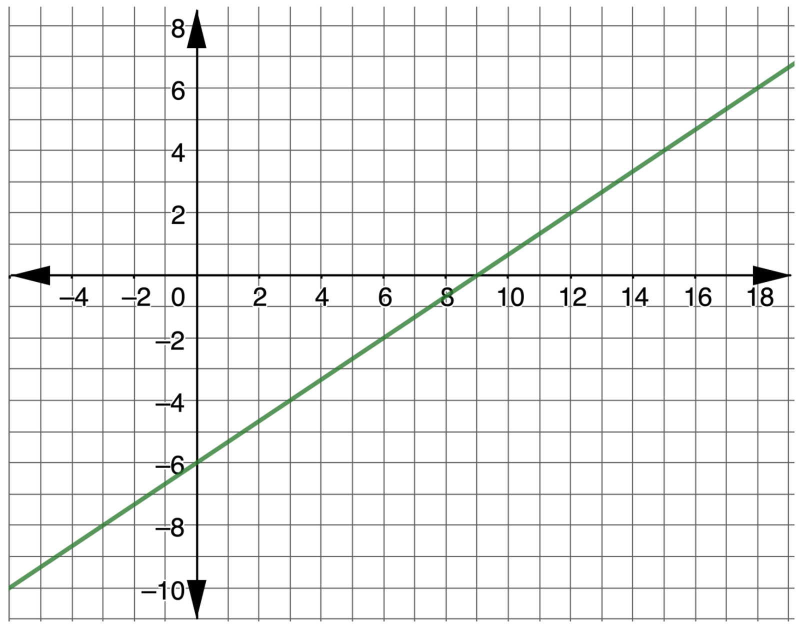
Type of Transformation	k-value
○ $f(kx)$ ○ $kf(x)$ ○ $f(x+k)$ ○ $f(x)+k$	○ -7 ○ 7

2. The table of values for $B(x)$ and $R(x)$ are shown, where $R(x)$ is a transformation of $B(x)$.

$y = B(x)$		$y = R(x)$	
x	y	x	y
-2	-15	-4	-15
-1	-11	-3	-11
0	-7	-2	-7
1	-3	-1	-3
2	1	0	1

Work with your partner to describe how the graph of $y = B(x)$ was transformed to create $y = R(x)$. Include the k-value in your description.

3. The graph of the function $W(x)$ is shown.



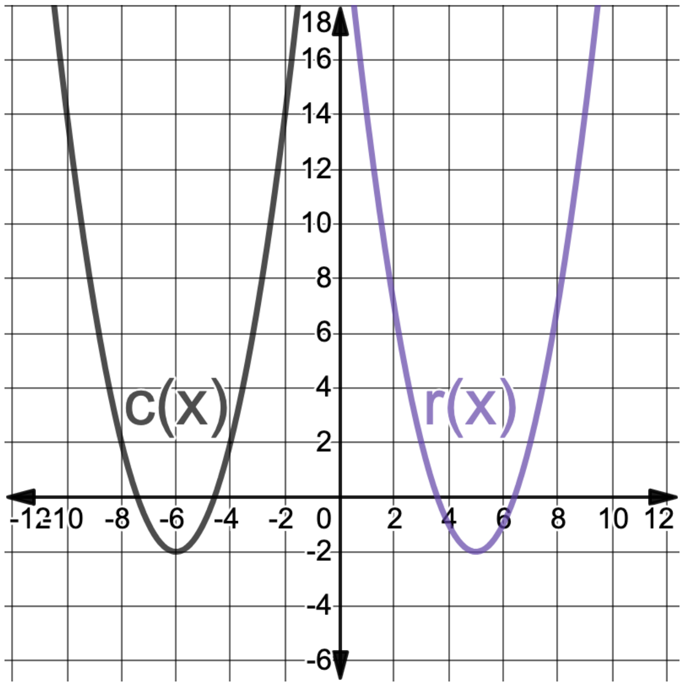
a. Write the function, $W(x)$, that represents the line.

b. Draw a line that is parallel to $W(x)$ with a y-intercept of 2. Label this line $M(x)$.

c. Describe how $W(x)$ can be shifted to coincide with $M(x)$.

d. Write the function $M(x)$ in terms of $W(x)$.

4. The functions $c(x)$ and $r(x)$ are shown on the graph.



a. Write each function in vertex form.

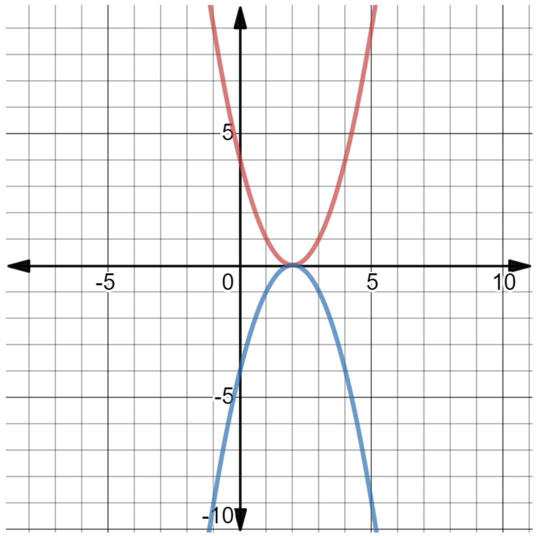
$c(x)$	$r(x)$

b. Describe how $c(x)$ was transformed to create $r(x)$.

c. Discuss with your partner how transformations of quadratic functions can be determined using the vertex form of the functions.

5. The functions $B(x)$ and $H(x)$ are shown on the graph. The table of values for $B(x)$ and $H(x)$ are shown, where $H(x)$ is a transformation of $B(x)$.

$y = B(x)$		$y = H(x)$	
x	y	x	y
-1	9	-1	-9
0	4	0	-4
1	1	1	-1
2	4	2	-4
3	9	0	-9

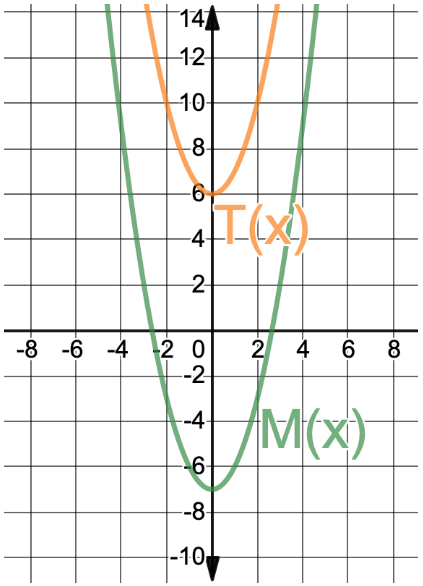


Work with your partner to complete the table to describe $H(x)$.

Type of Transformation	k-value
○ $B(kx)$ ○ $kB(x)$ ○ $B(x+k)$ ○ $B(x)+k$	

13.H1.3 Guided Practice: Describing Transformations

1. The functions $M(x)$ and $T(x)$ are shown on the graph, where $M(x)$ is a transformation of $T(x)$.



Write the transformation by completing the equation.

$$M(x) = T(x) + \_\_\_\_\_$$

2. The table of values for $B(x)$ and $R(x)$ are shown, where $R(x)$ is a transformation of $B(x)$. Write a description of the transformation.

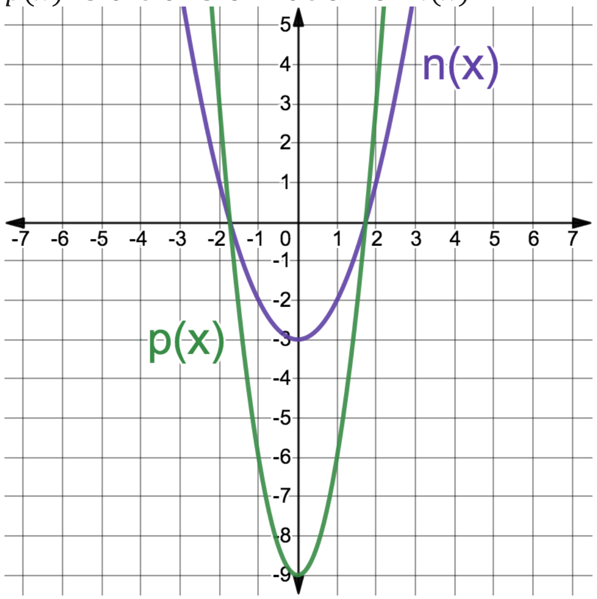
$y = B(x)$		$y = R(x)$	
x	y	x	y
9	10	1	10
7	8	-1	8
5	6	-3	6
3	8	-5	8
1	10	-7	10

13.H1.4 Your Turn!

1. The table of values for $f(x)$ and $g(x)$ are shown, where $g(x)$ is a transformation of $f(x)$. Write a description of the transformation.

$y = f(x)$		$y = g(x)$	
x	y	x	y
-2	-12	-2	-1
1	-6	1	5
2.5	-3	2.5	8
4	-6	4	5
7	-12	7	-1

2. The functions $n(x)$ and $p(x)$ are shown on the graph, where $p(x)$ is a transformation of $n(x)$.



What is the value of k such that $p(x) = kn(x)$?

13.H1.5 Wrap-Up: Reflection

Draw an **X** on each line to indicate how you feel about each learning target.

1. I can determine the k-value of a transformation.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

2. I can describe a transformation using words or function notation.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

Unit 13, Lesson 8: Identifying and Interpreting Parts of an Expression or Equation – Part 1

Benchmark

MA.912.AR.1.1 Identify and interpret parts of an equation or expression that represent a quantity in terms of a mathematical or real-world context, including viewing one or more of its parts as a single entity.

Learning Targets

- ☐ I can identify parts of an expression including factors, terms, constants, coefficients, and variables.
- ☐ I can interpret parts of an equation or expression that represent a quantity in terms of a mathematical or real-world context.